Literature Review

2.1 Technical & Domain Overview

Modern educational technology ecosystems require platforms that combine high availability, extensible architecture, and seamless multimedia streaming. Traditional monolithic learning architectures frequently bundle data access, presentation, and domain validation into tightly coupled assemblies, resulting in maintainability hurdles, challenging automated test cycles, and single-point-of-failure vulnerabilities under heavy loads.

EduHub addresses these historical limitations by applying modern enterprise software engineering paradigms through ASP.NET Core MVC. By dividing the system into distinct operational layers (Presentation, Business Logic, and Data Access), the solution ensures complete separation of concerns, testability via Dependency Injection, and readiness for cloud-native scaling.

2.2 Comparative Architecture & Pattern Evaluation

Feature / Architecture	Legacy Monolithic Systems	EduHub Advanced Implementation
Architectural Separation	Single-tier monolithic code with business logic embedded inside controllers and views.	Strict N-Tier Architecture isolating presentation (Web), domain logic (Services), and persistence (Data).
Database Abstraction	Ad-hoc SQL statements or coupled ORM calls scattered across UI code.	Generic Repository Pattern and Unit of Work abstracting Entity Framework Core, ensuring strict transaction atomicity.
Pricing & Discount Logic	Nested, hard-to-maintain conditional branches (if-else) for coupons and discounts.	Strategy Pattern enabling plug-and-play addition of seasonal discounts, fixed-amount vouchers, and percentage deductions.

Feature / Architecture	Legacy Monolithic Systems	EduHub Advanced Implementation
Course Lifecycle Events	Manual status polling and monolithic procedural calls for state changes.	Observer Pattern to asynchronously capture student milestone triggers, update progress counters, and trigger automated certification.
Video & Asset Storage	Local web server filesystem storage leading to bandwidth saturation and security leaks.	Azure Blob Storage with secure, expiring Shared Access Signature (SAS) tokens and CDN acceleration.
Credential Generation	Static HTML pages or pre-printed, non-verifiable template files.	Programmatically generated, vector-rendered PDF documents using iTextSharp with embedded cryptographic validation codes.

2.3 Academic & Industrial Standards Compliance

EduHub is structured to meet and exceed the grading benchmarks defined by the Digital Egypt Pioneers Initiative (DEPI):

- **SOLID Architecture: Strict adherence to Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, and Dependency Inversion principles.**
- **Enterprise Security Standards: Defense-in-depth security including anti-cross-site scripting (XSS), cross-site request forgery (CSRF) tokens, role/claim-based authorization, and robust SQL injection elimination via parameterization.**
- **Clean Code & Testability: Decoupled business components supporting dependency injection for automated unit and integration testing.**
- **Cloud-First Engineering: Cloud-ready architectural design verified through live deployment on Microsoft Azure infrastructure.**